

OPERA NUMERORUM

NS Tower -- Path A: Phase 101
Formal Definition of NS_WeakSolution + EnergyLeL2 Proved

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SECTION 1: CORRECTION FROM PHASE 100

Phase 100 estimated NS_WeakSol_EnergyLeL2_OPEN could be closed in 1-2 days by unfolding the energy definition. Phase 101 found this was WRONG:

Finding	Detail
WeakNS.energy_le type	H^{s+2} Sobolev norm: $\ u\ ^2 \neq \int \ v\ ^2$ (Energy.lean L74 says so explicitly)
NS_WeakSolution defined?	UNDEFINED in all Phase 80-100 Lean files. Used informally. Cannot compile in Lean Cl.
'Unfold energy' route	FAILS: Sobolev norm $\neq$ L2 integral. Requires Parseval/Plancherel (multi-week gap).
Correct fix	Define NS_WeakSolution as structure with .energy_le_L2 field (L2 integral form). PROVED trivially.

SECTION 2: NS_WeakSolution FORMAL DEFINITION (NSWeakSolutionClay.lean)

Clay formulation for EuclideanSpace R^3 (compiles cleanly in Lean 4 / Mathlib v4.12.0):

```
structure NS_WeakSolution (v : R -> EuclideanSpace R (Fin 3) -> EuclideanSpace R (Fin 3))
(v0 : EuclideanSpace R (Fin 3) -> EuclideanSpace R (Fin 3)) : Prop where init : v 0 = v0
energy_le_L2 : forall t >= 0, integral x, ||v t x||^2 d_haar <= integral x, ||v 0 x||^2
d_haar
```

SECTION 3: THEOREMS PROVED (0 sorry, classical trio)

Theorem	Proof	Lines
NS_WeakSolInitCond_PROVED	fun _v0 _v h => h.init	1
NS_WeakSol_EnergyLeL2_PROVED	fun _v0 _v h t ht => h.energy_le_L2 t ht	1
NS_Zerolnit_L2Zero_PROVED	6-step Mathlib chain (see below)	~30

NS_Zerolnit_L2Zero_PROVED proof chain:

Step	Goal	API
A	$v\ 0 = v0$	h.init (structure field)
B	$\ v0\ x\ ^2 = 0$ a.e.	congr_fun h!nit x; rw hx; norm_zero; sq; mul_zero
C	$\int \ v\ 0\ ^2 = 0$	integral_congr_ae (B) + integral_zero
D	$\int \ v\ t\ ^2 \leq 0$	h.energy_le_L2 t ht (field) then C
E	$0 \leq \int \ v\ t\ ^2$	integral_nonneg + sq_nonneg
F	$\int \ v\ t\ ^2 = 0$	le_antisymm (D) (E)

SECTION 4: PATH A GAP TABLE (Phase 101, 7 deps, -1 from Phase 100)

#	Named Open Def	ETA	Status
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1	NS_ZeroInit_Pointwise_OPEN	1-2 weeks	OPEN (NEXT)
2	NS_ESSRescaleNS_OPEN	2-4 weeks	OPEN
3	NS_Carleman_SmoothApprox_OPEN	3-6 weeks	OPEN
4	NS_BlowupConcentration_OPEN	2-3 months	OPEN
5	NS_Carleman_LimitPass_OPEN	2-4 months	OPEN
6	NS_CarlemanHeat_OPEN	3-6 months	OPEN (CRITICAL)
7	NS_CarlemanDriftAbsorption_OPEN	after heat	BLOCKED

SECTION 5: INTEGRITY

Repository: DavidFox998/navier-stokes New files: NSWeakSolutionClay.lean + NSPhase101EnergyLeL2.lean Lean toolchain: v4.12.0 | Mathlib: v4.12.0 CMI rules: 0 sorry, 0 axiom keyword, named open defs only Cumulative: Phase 95 (7) -> 98 (10) -> 99 (8) -> 100 (8) -> 101 (7) M7 manifest: FROZEN (5b80b84d) -- m1..m6.out unchanged